AI STUDY COMPANION · RETRIEVAL PRACTICE

CogniGuide

Upload your notes. Get questions written from them. Answer, get graded against your own material, and keep reviewing what you missed.

Zachary Zhang · Hugo Wong

WHY COGNIGUIDE

Re-reading notes feels productive but builds little memory; **testing yourself** (active recall) works far better. CogniGuide turns handwritten or digital notes into short-answer quizzes, so every session becomes self-testing on **your own material**.

Passive re-reading
Familiarity feels like understanding; gaps stay hidden.

Active recall with CogniGuide
Scores and feedback reveal what you have not mastered.

DESIGN PRINCIPLES

Grounded in your notes
RAG: questions and grading are based on the original note chunks.

Level-aware
Hong Kong levels S1–S6; questions and grading adapt to the level.

Learn from mistakes
Scores below 60 are logged and reviewed until mastered.

Cloud or fully local
Runs on your own machine, no login. Use Bailian cloud models, or Ollama so notes never leave it.

HOW IT WORKS: FROM NOTES TO MASTERY

01 · UPLOAD
Add notes
PDFs or photos of handwritten notes, sorted into subject folders and processed in the background.

`PDF · PNG · JPG · WEBP`

02 · TRANSCRIBE
Image → text
Each page is rendered to an image; a vision model transcribes text and math into Markdown + LaTeX.

`CLOUD qwen3-vl-plus`
`LOCAL qwen2.5vl:3b`

03 · INDEX
Chunk & embed
Text is split into ~800-character overlapping chunks, embedded and stored in LanceDB for semantic search.

`CLOUD text-embedding-v4`
`LOCAL bge-m3`

04 · GENERATE
Write questions
An LLM writes five short-answer questions per material, tuned to the student's level.

`CLOUD qwen-plus`
`LOCAL qwen3:4b-instruct`

05 · GRADE
Score & feedback
Answers are scored 0–100 against the source notes, with correct points, missed points and feedback.

`CLOUD qwen-plus`
`LOCAL qwen3:4b-instruct`

↺ **Feedback loop:** any answer scoring below 60 is saved to the Mistakes log → review rounds of five → a later score of 60 or more marks it **mastered**. Quick Quiz mixes questions from all subjects, optionally focused on a topic via semantic search.

SYSTEM ARCHITECTURE

FRONTEND
Next.js 16 · React 19
Subjects · Practice · Quick Quiz · Mistakes · Search · Level picker · KaTeX

API & SERVICES
Next.js route handlers
`/api/grade` `/api/search` `/api/quick-quiz` `/api/mistakes`

jobs queue · **extract** · **rag** · **quiz**

STORAGE
Local data directory
LanceDB vectors · extract.md · questions-<level>.json · attempts · mistakes

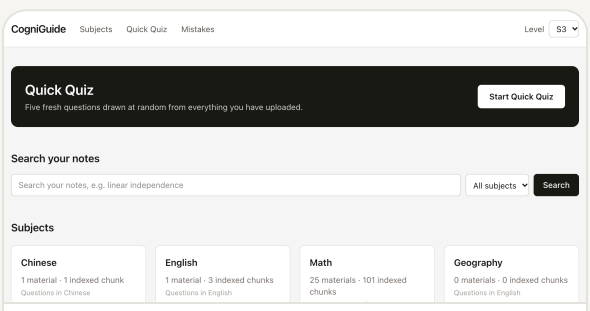
Transcripts and questions are cached.

AI MODELS · SWITCHABLE
MODEL_PROFILE = cloud | local
Same OpenAI-compatible API for both

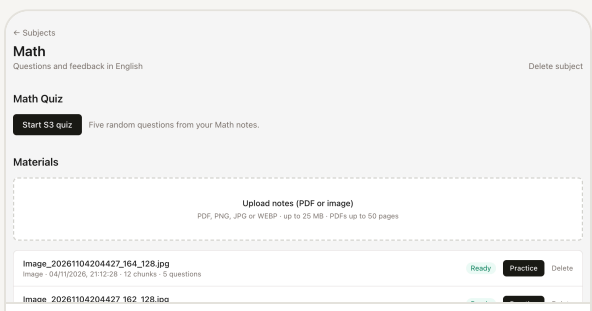
	Cloud · Bailian	Local · Ollama
Vision	`qwen3-vl-plus`	`qwen2.5vl:3b`
Embed	`text-embedding-v4`	`bge-m3`
Chat	`qwen-plus`	`qwen3:4b-instruct`

Local: ~6.4 GB of models, works fully offline. Switching profiles needs one index rebuild.

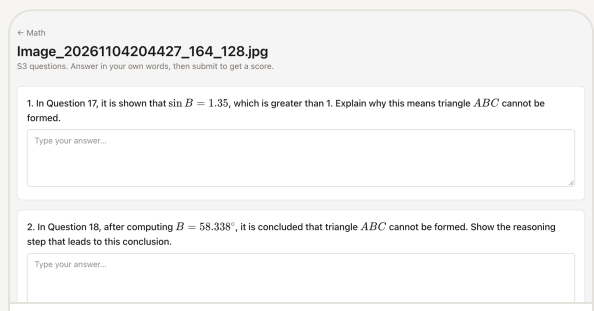
THE APP IN ACTION



Home. Quick Quiz, semantic note search and subject folders with indexed chunks.



Subject page. Upload notes, track processing status and start a five-question quiz.



Practice. Questions generated from a handwritten math note, rendered with LaTeX.

DEMO VIDEO ON YOUTUBE
`youtu.be/5Exw-_zTLdY`
Scan to watch the walkthrough: each member presents the code they built.

SOURCE CODE ON GITHUB
`github.com/ZZZ-zac/CogniGuide`
Runs on Windows, macOS and Ubuntu with Node.js 22, using Bailian cloud models or local Ollama models.

AUTHORS
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